ON THE IDEAL BILINEAR AND BIQUADRATIC DIGITAL FILTER

Yuriy Ivantsov
Sofia, Bulgaria. 16 April 2025
Revised 15 March 2026

Abstract. A method is presented for computing the coefficients of a second-order digital biquadratic filter that matches, as closely as possible, the characteristics of the analogue prototype. It is shown that, by first determining the coefficients of the first-order system, one may extract certain information that makes the second-order problem tractable. A solution in general form is obtained, expressed entirely in algebraic functions and valid in the case where the cutoff frequency tends to infinity.

1. INTRODUCTION

When a system is designed in the analogue domain and must subsequently be realised digitally, the central difficulty is the passage from the continuous s-plane to the discrete z-plane. The classical approach, known as the bilinear transform, achieves this passage by a direct substitution of variables; a simplicity that accounts for its widespread use. The method is not, however, without defect. At high cutoff frequencies, the magnitude response departs perceptibly from that of the analogue prototype and suffers compression near the Nyquist frequency. Various alternative procedures have been proposed with the aim of correcting these deficiencies, but they rely on system-dependent computation, involve arithmetic of considerable complexity, and neglect the phase response. The purpose of the present paper is to derive a simple and accurate solution applicable to any given system, and simple results, as a rule, always follow from simple derivations. In Section 2, we exploit the known location of the zeros of a discrete high-pass filter to resolve the first-order problem in closed form. Then, in Section 3, we apply the same reasoning to the second-order problem, where a quantity emerging from the first-order solution plays an essential role.

2. THE FIRST-ORDER FILTER

2.1. The high-pass case

A first-order analogue high-pass filter, with cutoff frequency f_c, has the magnitude response

$$\Omega_A(x) = \frac{x}{\sqrt{x^2 + f_c^2}} \tag{2.1}$$

A digital bilinear filter, with sampling frequency f_s, has the discrete transfer function

$$H(z) = \frac{a_0 + a_1 z^{-1}}{b_0 + b_1 z^{-1}} = A \cdot \frac{1 + \alpha z^{-1}}{1 + \beta z^{-1}} \tag{2.2}$$

with magnitude response

$$\Omega_D(x) = A \cdot \sqrt{\frac{\alpha^2 + 2\alpha \cos\left(\frac{2\pi x}{f_s}\right) + 1}{\beta^2 + 2\beta \cos\left(\frac{2\pi x}{f_s}\right) + 1}} \tag{2.3}$$

Let us introduce the ratio $\omega = f_s/f_c$. We require the digital filter to match its analogue prototype at $x = 0$ both in gain and in slope.

$$\left.\begin{aligned} \mathrm{D}_x^0 \Omega_D(0) &= \mathrm{D}_x^0 \Omega_A(0) \\ \mathrm{D}_x^1 \Omega_D(0) &= \mathrm{D}_x^1 \Omega_A(0) \end{aligned}\right\} \implies \begin{cases} \alpha = -1 \\ A = \left(1 + \beta\right)\left(\frac{\omega}{2\pi}\right) \end{cases} \tag{2.4}$$

In order to find β, we impose a gain matching condition at f_s/σ, where σ is a free parameter.

$$\begin{aligned} &\Omega_D\left(\frac{f_s}{2}\right) = \Omega_A\left(\frac{f_s}{\sigma}\right) \\ \implies \quad &\beta = \frac{\pi - \nu}{\pi + \nu}, \qquad \text{with } \nu = \sqrt{\omega^2 + \sigma^2} \end{aligned} \tag{2.5}$$

2.2. Further filter types of the first-order

The result of the preceding section admits a natural generalisation. Let us set

$$\varphi(x) = \frac{\pi - \sqrt{x^2 + \sigma^2}}{\pi + \sqrt{x^2 + \sigma^2}} \tag{2.6}$$

where a straightforward computation shows that

$$\lim_{x\to 0}\varphi(x)=\frac{\pi-\sigma}{\pi+\sigma}\quad\text{and}\quad\lim_{x\to\infty}\varphi(x)=-1 \tag{2.7}$$

With the aid of this function, distinct first-order transfer functions may be written in the unified form

$$H(z)=G\cdot\left(1+\beta\right)\left(\frac{1+\alpha z^{-1}}{1+\beta z^{-1}}\right) \tag{2.8}$$

Table 2.1 records the values of α, β and G for each standard filter type. The high-pass filter appears there as a special case.

Table 2.1.
First-order coefficients for various systems.

	α	β	G
High-pass	$\varphi(\infty)$	$\varphi(\omega)$	$\frac{\omega}{2\pi}$
Low-pass	$\varphi(0)$	$\varphi(\omega)$	$\frac{1}{1+\alpha}$
All-pass	$\varphi(\omega)^{-1}$	$\varphi(\omega)$	$\frac{1}{1+\alpha}$
High-shelf	$\varphi\left(\omega g^{1/2}\right)$	$\varphi\left(\omega g^{-1/2}\right)$	$\frac{1}{1+\alpha}$
Low-shelf	$\varphi\left(\omega g^{-1/2}\right)$	$\varphi\left(\omega g^{1/2}\right)$	$\frac{g}{1+\alpha}$

It is instructive to examine the frequency mapping function. The analogue magnitude response attains the value $1/\sqrt{2}$ at the cutoff frequency f_c; we require the digital response to satisfy the same condition.

$$\begin{aligned}\Omega_D(x)&=\frac{1}{\sqrt{2}}\\ \Longrightarrow\quad F(\omega)&=\frac{f_s}{2\pi}\arccos\left(\frac{\omega^2-\pi^2-\sigma^2}{\omega^2+\pi^2-\sigma^2}\right)\end{aligned} \tag{2.9}$$

As shown in Figure 2.1, the ideal magnitude response match is obtained when σ belongs to the interval $\left[2,\pi\sqrt{2/3}\right]$. This is precisely the range in which $F(\omega)=f_c$. On the other hand, the ideal phase response match is achieved when $\sigma=0$. Since $F(\omega)$ causes a slight shift in the cutoff frequency f_c, it may on occasion be desirable to reverse this effect, either by correcting ω directly, according to

$$F^{-1}(\omega)=\frac{f_s}{\sqrt{\sigma^2+\pi^2\operatorname{ctg}^2(\pi/\omega)}} \tag{2.10}$$

or by adjusting the free parameter σ according to

$$F^{-1}(\sigma)=\sqrt{\omega^2-\pi^2\operatorname{ctg}^2(\pi/\omega)} \tag{2.11}$$

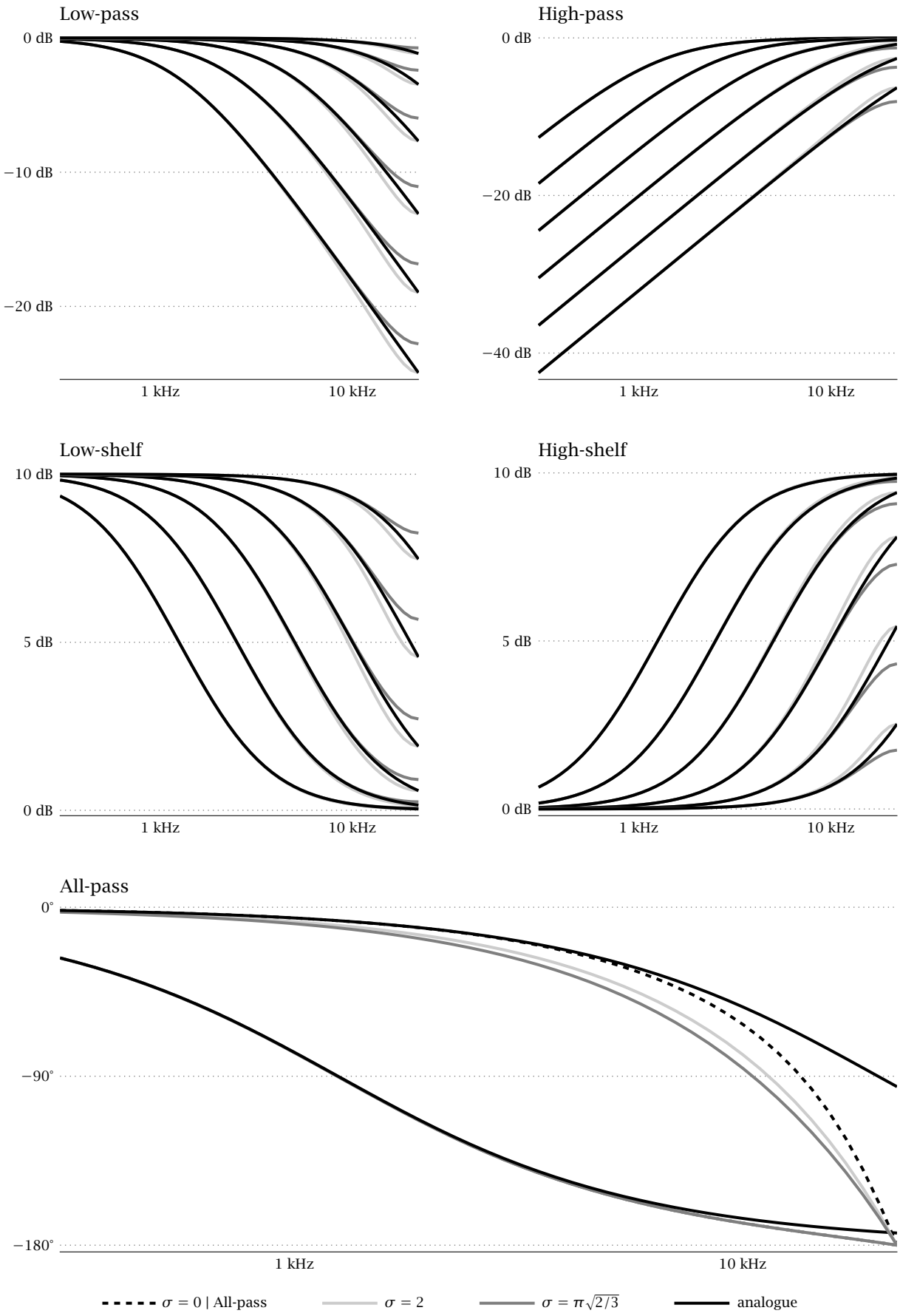


Figure 2.1. Different first-order filters with $f_c = [1250, 2500, 5000, 10\,000, 20\,000, 40\,000]$ Hz, $f_s = 44.1$ kHz, $g = 10$ dB.

3. THE SECOND-ORDER FILTER

3.1. The high-pass case

A second-order analogue high-pass filter, with damping ratio ζ, has the magnitude response

$$\Omega_A(x) = \frac{x^2}{\sqrt{x^4 + 2x^2 f_c^2 (2\zeta^2 - 1) + f_c^4}} \tag{3.1}$$

A digital biquadratic filter has the discrete transfer function

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2}}{b_0 + b_1 z^{-1} + b_2 z^{-2}} = A \cdot \frac{1 + \alpha_1 z^{-1} + \alpha_2 z^{-2}}{1 + \beta_1 z^{-1} + \beta_2 z^{-2}} \tag{3.2}$$

with magnitude response

$$\Omega_D(x) = A \cdot \sqrt{\frac{4\alpha_2 \cos^2\left(\frac{2\pi x}{f_s}\right) + 2\alpha_1(\alpha_2 + 1)\cos\left(\frac{2\pi x}{f_s}\right) + \alpha_1^2 + (\alpha_2 - 1)^2}{4\beta_2 \cos^2\left(\frac{2\pi x}{f_s}\right) + 2\beta_1\,(\beta_2 + 1)\cos\left(\frac{2\pi x}{f_s}\right) + \beta_1^2 + (\beta_2 - 1)^2}} \tag{3.3}$$

Proceeding as before, we require the gain, its first derivative and its second derivative at $x = 0$ all to match the analogue prototype, yielding the system of equations

$$\left.\begin{aligned} \mathrm{D}_x^0\Omega_D(0) &= \mathrm{D}_x^0\Omega_A(0) \\ \mathrm{D}_x^1\Omega_D(0) &= \mathrm{D}_x^1\Omega_A(0) \\ \mathrm{D}_x^2\Omega_D(0) &= \mathrm{D}_x^2\Omega_A(0) \end{aligned}\right\} \implies \begin{cases} \alpha_1 = -2 \\ \alpha_2 = 1 \\ A = \left(1 + \beta_1 + \beta_2\right)\left(\frac{\omega}{2\pi}\right)^2 \end{cases} \tag{3.4}$$

To find β_1, we employ the same gain matching condition as in the first-order case (2.5).

$$\begin{gathered} \Omega_D\left(\frac{f_s}{2}\right) = \Omega_A\left(\frac{f_s}{\sigma}\right) \\ \implies \beta_1 = \frac{\pi^2 - \nu}{\pi^2 + \nu} \cdot (1 + \beta_2)\,, \qquad \text{with } \nu = \sqrt{\omega^4 + 2\omega^2\sigma^2(2\zeta^2 - 1) + \sigma^4} \end{gathered} \tag{3.5}$$

To find β_2, we make use of the frequency mapping function F introduced in (2.9). We require the magnitude response at the frequency $F(\omega)$ to equal the resonance value $Q = 1/2\zeta$.

$$\begin{gathered} (\Omega_D \circ F)(\omega) = \frac{1}{2\zeta} \\ \implies \beta_2 = \frac{\pi^2 - \pi\sqrt{2}\sqrt{\nu + \kappa} + \nu}{\pi^2 + \pi\sqrt{2}\sqrt{\nu + \kappa} + \nu}\,, \qquad \text{with } \kappa = \omega^2\left(2\zeta^2 - 1\right) + \sigma^2 \end{gathered} \tag{3.6}$$

3.2. Further filter types of the second-order

The construction of the preceding section generalises in a manner entirely analogous to the first-order case. Let us introduce the auxiliary quantities

$$\begin{aligned}
\nu(x,y) &= \sqrt{x^4 + 2x^2\sigma^2(2y^2-1) + \sigma^4} \\
\kappa(x,y) &= x^2\big(2y^2-1\big) + \sigma^2 \\
\varphi_1(x,y) &= \frac{2\pi^2 - 2\nu(x,y)}{\pi^2 + \pi\sqrt{2}\sqrt{\nu(x,y)+\kappa(x,y)} + \nu(x,y)} \\
\varphi_2(x,y) &= \frac{\pi^2 - \pi\sqrt{2}\sqrt{\nu(x,y)+\kappa(x,y)} + \nu(x,y)}{\pi^2 + \pi\sqrt{2}\sqrt{\nu(x,y)+\kappa(x,y)} + \nu(x,y)}
\end{aligned} \tag{3.7}$$

A second-order transfer function may then be written in the form

$$H(z) = G \cdot \Big(1 + \beta_1 + \beta_2\Big)\left(\frac{1 + \alpha_1 z^{-1} + \alpha_2 z^{-2}}{1 + \beta_1 z^{-1} + \beta_2 z^{-2}}\right) \tag{3.8}$$

with the coefficients collected in Table 3.1.

Table 3.1.
Second-order coefficients for various systems; with $\varphi(x)$ from Section 2.2.

	α_1	α_2	β_1	β_2	G
High-pass	$\varphi(\infty) + \varphi(\infty)$	$\varphi(\infty) \cdot \varphi(\infty)$	$\varphi_1(\omega,\zeta)$	$\varphi_2(\omega,\zeta)$	$\dfrac{\omega^2}{4\pi^2}$
Band-pass	$\varphi(0) + \varphi(\infty)$	$\varphi(0) \cdot \varphi(\infty)$	$\varphi_1(\omega,\zeta)$	$\varphi_2(\omega,\zeta)$	$\dfrac{\omega}{2\pi} \cdot \dfrac{2\zeta}{2+\alpha_1}$
Low-pass	$\varphi(0) + \varphi(0)$	$\varphi(0) \cdot \varphi(0)$	$\varphi_1(\omega,\zeta)$	$\varphi_2(\omega,\zeta)$	$\dfrac{1}{1+\alpha_1+\alpha_2}$
All-pass	$\varphi_1\varphi_2^{-1}(\omega,\zeta)$	$\varphi_2^{-1}(\omega,\zeta)$	$\varphi_1(\omega,\zeta)$	$\varphi_2(\omega,\zeta)$	$\dfrac{1}{1+\alpha_1+\alpha_2}$
Band-stop	$\varphi_1(\omega,0)$	$\varphi_2(\omega,0)$	$\varphi_1(\omega,\zeta)$	$\varphi_2(\omega,\zeta)$	$\dfrac{1}{1+\alpha_1+\alpha_2}$
Band-shelf	$\varphi_1\big(\omega,\zeta g^{1/2}\big)$	$\varphi_2\big(\omega,\zeta g^{1/2}\big)$	$\varphi_1\big(\omega,\zeta g^{-1/2}\big)$	$\varphi_2\big(\omega,\zeta g^{-1/2}\big)$	$\dfrac{1}{1+\alpha_1+\alpha_2}$
High-shelf	$\varphi_1\big(\omega g^{1/4},\zeta\big)$	$\varphi_2\big(\omega g^{1/4},\zeta\big)$	$\varphi_1\big(\omega g^{-1/4},\zeta\big)$	$\varphi_2\big(\omega g^{-1/4},\zeta\big)$	$\dfrac{1}{1+\alpha_1+\alpha_2}$
Low-shelf	$\varphi_1\big(\omega g^{-1/4},\zeta\big)$	$\varphi_2\big(\omega g^{-1/4},\zeta\big)$	$\varphi_1\big(\omega g^{1/4},\zeta\big)$	$\varphi_2\big(\omega g^{1/4},\zeta\big)$	$\dfrac{g}{1+\alpha_1+\alpha_2}$

We remark that in the band stop case, the feedforward coefficients α_1 and α_2 may be simplified at the cost of losing the minimum phase property when $\omega < \sigma$. The simplified expressions are

$$\alpha_1 = -2 \cdot \frac{\omega^2 - \pi^2 - \sigma^2}{\omega^2 + \pi^2 - \sigma^2} \quad \text{and} \quad \alpha_2 = 1 \tag{3.9}$$

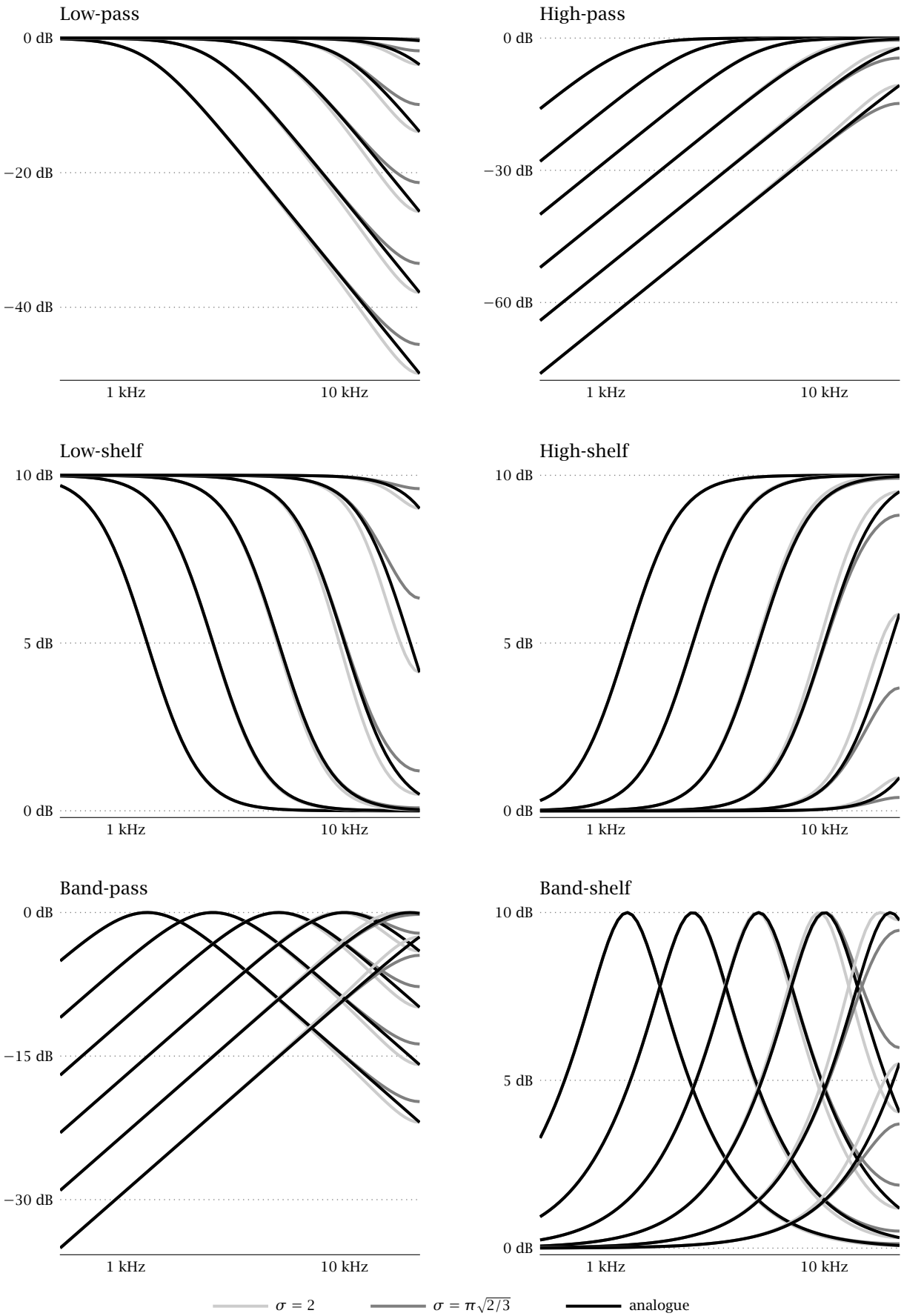


Figure 3.2. Different second-order filters with $f_c = [1250, 2500, 5000, 10\,000, 20\,000, 40\,000]$ Hz, $f_s = 44.1$ kHz, $g = 10$ dB, $\zeta = 1/\sqrt{2}$.

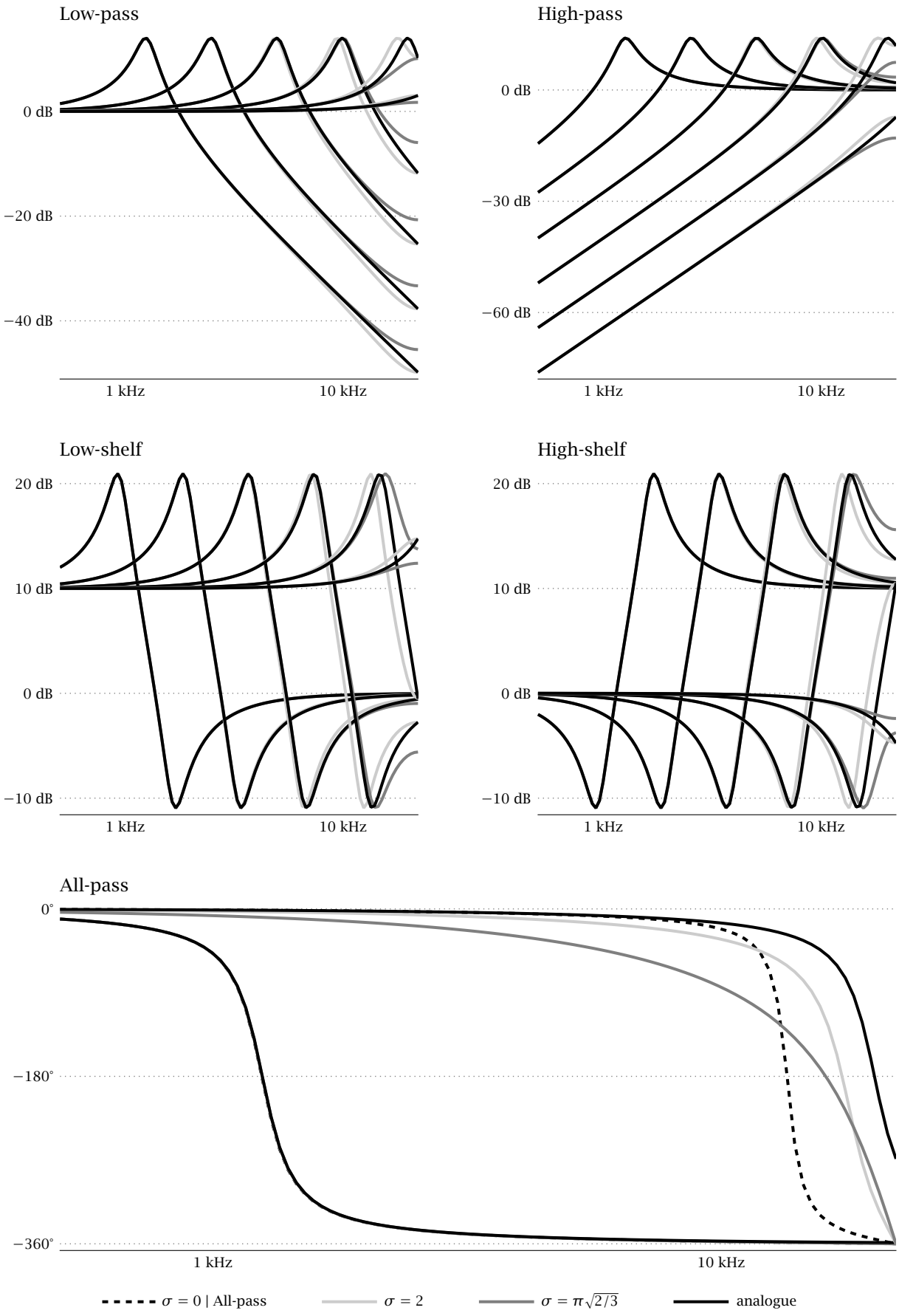


Figure 3.3. Different second-order filters with $f_c = [1250, 2500, 5000, 10\,000, 20\,000, 40\,000]$ Hz, $f_s = 44.1$ kHz, $g = 10$ dB, $\zeta = 1/10$.

4. FILTERS OF HIGHER ORDERS

4.1. The general form

We now consider filters of arbitrary order N. The fundamental theorem of algebra asserts that every real polynomial of degree N admits a factorisation into real linear and quadratic factors:

$$\sum_{n=0}^{N} a_n s^n = \underbrace{(a_{0,0} + a_{0,1} s)}_{\text{if } N \text{ is odd}} \cdot \prod_{n=1}^{\lfloor N/2 \rfloor} \sum_{m=0}^{2} a_{n,m} s^m , \qquad \text{with } a_n, a_{n,m} \in \mathbb{R} \tag{4.1}$$

A first-order s-plane polynomial is mapped to the z-plane by the substitution

$$\begin{aligned} as + 1 \quad &\circ\!\!-\!\!\bullet \quad \frac{1 + cz^{-1}}{1 + c} \\ s \quad &\circ\!\!-\!\!\bullet \quad \left(1 - z^{-1}\right) \cdot \frac{\omega}{2\pi} \end{aligned} \tag{4.2}$$

where

$$c = \begin{cases} \varphi(\omega a) & \text{for } a \geqslant 0, \\ \varphi(\omega a)^{-1} & \text{for } a \leqslant 0. \end{cases} \tag{4.3}$$

For a second-order s-plane polynomial with real roots, the mapping (4.2) is applied separately to each linear factor. When the roots are complex, however, a different mapping is required, namely

$$as^2 + bs + 1 \quad \circ\!\!-\!\!\bullet \quad \frac{1 + c_1 z^{-1} + c_2 z^{-2}}{1 + c_1 + c_2} \tag{4.4}$$

where

$$c_{\{1,2\}} = \begin{cases} \left\{\varphi_1(x,y), \varphi_2(x,y)\right\} & \text{for } b \geqslant 0, \\ \left\{\varphi_1(x,y)\varphi_2(x,y)^{-1}, \varphi_2(x,y)^{-1}\right\} & \text{for } b \leqslant 0. \end{cases} \tag{4.5}$$

and

$$x = \omega\sqrt{a} \quad \text{and} \quad y = \frac{b}{2\sqrt{a}} \tag{4.6}$$

For the degenerate cases in which a, or a and b, vanish, the limiting forms are taken in the obvious manner

$$bs + 1 = (as + 1)(bs + 1) \quad \text{and} \quad 1 = (as + 1)(as + 1), \qquad \text{with } a = 0 \tag{4.7}$$

4.2. A four-pole example

By way of illustration, let us apply the general theory to a specific example. Consider a four-pole closed-loop low-pass filter with feedback coefficient K. Its continuous transfer function is

$$H(s) = \frac{1 + K^4}{(1+s)^4 + K^4} \tag{4.8}$$

Then, with

$$a = 0, \qquad a_{\{1,2\}} = \frac{1}{\sqrt{K^2 \pm \sqrt{2}K + 1}}, \qquad b_{\{1,2\}} = 2 \pm \sqrt{2}K \tag{4.9}$$

the transfer function may be factored and each factor mapped to the z-plane using the mappings established above:

$$\begin{array}{rcccl} H(s) = & \Big(as+1\Big)^4 & \times & \left(\prod_{n=1}^{2}\Big(a_n^2 s^2 + a_n^2 b_n s + 1\Big)\right)^{-1} \\ & \Big\downarrow & & \Big\downarrow \\ H(z) = & \left(\dfrac{1+\varphi(\omega a)z^{-1}}{1+\varphi(\omega a)}\right)^4 & \times & \left(\prod_{n=1}^{2}\dfrac{1+\sum_{m=1}^{2}\varphi_m\Big(\omega a_n, \frac{1}{2}a_n b_n\Big)z^{-m}}{1+\sum_{m=1}^{2}\varphi_m\Big(\omega a_n, \frac{1}{2}a_n b_n\Big)}\right)^{-1} \end{array} \tag{4.10}$$

5. CONCLUDING REMARKS

For elementary systems, that is, for low-pass, high-pass and band-pass filters whose zeros in the continuous domain lie at 0 or at $\pm\infty$, the zeros of the discrete transfer function reduce to constants.

The free parameter σ may be fine-tuned according to preference. Setting $\sigma = 0$ substantially simplifies the computation of the coefficients, at the cost of introducing frequency cramping at the Nyquist frequency. For small values of f_c, this drawback is negligible in practice.

The frequency compensation mapping (2.10) and (2.11) may be employed whenever it is desired that the digital filter possess the same cutoff frequency as the analogue prototype.

If the poles of a continuous system lie in the right half of the s-plane, there exists a specific cutoff frequency f_c at which the mappings (4.3) and (4.5) entail a division by zero.

We observe, finally, that the mappings φ_1 and φ_2 defined in (3.7) share a common denominator d. It is a noteworthy algebraic fact that the sum $1 + \varphi_1 + \varphi_2$ may be written simply as $4\pi^2/d$. This identity, far from being a coincidence, reflects the structure of the mapping and may be exploited to simplify the computation of the gain coefficient G for each filter type.